GoHydro algorithm

Simplified production scheduling problem for PoC

Based on the general production scheduling problem, a subset of features is included in the proof-of-concept for the GoHydro algorithm. This is necessary to be able to test imporant basic properties of the algorithm within the scope of this project. The following simplifications are made in this version.

- The only commodity traded is energy.
- The price is given, and there is no maximum limit for selling to the market.
- All generators are in the same price area.
- There are no pumps in the system.
- There are no thermal units in the system.
- There are no power lines in the system.
- The minimum production of each generating unit is 0.
- There are no startup costs or ramping restrictions in the system.
- There are no minimum pressure restrictions in tunnels.
- There are no gates inside tunnels.
- All scenarios are assumed to have equal probability
- Water level in each reservoir is assumed to be constant for energy production.
- There is no time delay in rivers.

These simplifications result in the following formulation.

Sets

Symbol	Comment
$t \in T$	Timesteps
$s \in S$	Scenarios

Symbol	Comment
$g \in G$	Hydropower generating units
$m \in M_g$	PQ segments for generator g
$r \in R$	Reservoirs
$n \in N$	Tunnels
$e \in E$	Rivers

Subsets

Symbol	Comment
r_g^{in}	Reservoir discharging water to generating unit g
r_g^{out}	Reservoir receiving water from generating unit g
r_n^{start}	Reservoir at the start of tunnel n
r_n^{end}	Reservoir at the end of tunnel n
G_r^{in}	Generating units discharging water to reservoir r
G_r^{out}	Generating unit receiving water from reservoir r
E_r^{in}	Rivers discharging water to reservoir r
E_r^{out}	Rivers receiving water from reservoir r
N_r^{in}	Tunnels discharging water to reservoir r
N_r^{out}	Tunnels receiving water from reservoir r
t_{end}	Last timestep

Parameters

Symbol	Comment
$\eta_{g,m}^{max}$	PQ-curve coefficient for energy conversion efficiency at generator g for pq-segment m

Symbol	Comment
$Q_{g,m}^{min}$	Minimum discharge for generator g and pq-segment m in timestep t
$Q_{g,m}^{max}$	Maximum discharge for generator g and pq-segment m in timestep t
$Q_{e,t}^{min}$	Minimum flow in river e in timestep t
$Q_{e,t}^{max}$	Maximum flow in river e in timestep t
Q_{n}^{max}	Maximum flow in tunnel n
$V_{r,t}^{min}$	Minimum storage volume for reservoir r in timestep t
$V_{r,t}^{max}$	Maximum storage volume for reservoir r in timestep t
V_r^0	Initial volume for reservoir r
$C_{t,s}^{market}$	Price in market in timestep t and scenario s
$C_{e,t,s}^{river}$	Cost for discharge in river r in timestep t and scenario s
$I_{r,t,s}$	Natural inflow to reservoir r in hour t and scenario s
Ω_r	Expected income per unit of water in the future for reservoir r
τ_t	Timestep length in hours for timestep t
$\alpha_{n,t}^{max}$	Friction loss factor in tunnel n in timestep t
$H_r(v_r)$	Relationship between head and volume in reservoir r

Variables

Symbol	Comment
z	Objective value
$p_{g,t,s}^{gen}$	Production at generating unit g in hour t and scenario s
$q_{g,m,t,s}^{gen}$	Discharge in pq segment m at generator g in hour t and scenario s
$q_{e,t,s}^{river}$	Flow in river e in hour t and scenario s

Symbol	Comment
$q_{n,t,s}^{tunnel}$	Flow in tunnel n in hour t and scenario s
$v_{r,t,s}$	Stored volume in reservoir r at the end of hour t in scenario s

Objective function

The objective is to maximize the expected value of remaining water plus income from selling power to the market.

$$maxz = \sum_{s \in S}(\sum_{r \in R}(\Omega_r v_{r,t_{end},s}) + \sum_{t \in T} \tau_t (C_{t,s}^{market} \sum_{g \in G} p_{g,t,s}^{gen} - C_{t,s}^{river} \sum_{e \in E} q_{e,t,s}^{river})) \quad (1)$$

Constraints

Hydropower generating units

The power produced by each generator is equal to sum of energy conversion in all PQ-segments. This is found by multiplying the coefficient of each PQ-segment with the discharge in that segment.

$$p_{g,t,s}^{gen} = \sum_{m \in M_g} (\eta_{g,m} q_{g,m,t,s}^{gen}) \quad \forall g \in G, t \in T, s \in S \quad (2)$$

The discharge in each pq-segment must be between min and max limits.

$$q_{g,m,t,s}^{gen} >= 0 \quad \forall g \in G, m \in M_g, t \in T, s \in S \quad (3)$$

$$q_{g,m,t,s}^{gen} <= Q_{g,m}^{max} \quad \forall g \in G, m \in M_g, t \in T, s \in S \quad (4)$$

Reservoirs

The volume in the reservoir is equal to the previous volume plus inflow and water received from upstream generators and rivers minus water delivered to downstream generators and rivers.

$$v_{r,t,s} = v_{r,t-1,s} + \tau_t (I_{r,t,s} + \sum_{g' \in G_r^{in}} q_{g',t,s}^{gen} + \sum_{e' \in E_r^{in}} q_{e',t,s}^{river} - \sum_{g'' \in G_r^{out}} q_{g'',t,s}^{gen} - \sum_{e'' \in E_r^{out}} q_{e'',t,s}^{river}) \quad \forall r \in R, t \in T, s \in S \quad (5)$$

The volume in each reservoir must be within minimum and maximum limits.

$$v_{r,t,s} >= V_{r,t}^{min} \quad \forall r \in R, t \in T, s \in S \quad (6)$$

$$v_{r,t,s} <= V_{r,t}^{max} \quad \forall r \in R, t \in T, s \in S \quad (7)$$

Rivers

The flow in each river must be between minimum and maximum discharge capacity.

$$q_{e,t,s}^{river} >= Q_{e,t,s}^{min} \quad \forall e \in E, t \in T, s \in S \quad (8)$$

$$q_{e,t,s}^{river} <= Q_{e,t,s}^{max} \quad \forall e \in E, t \in T, s \in S \quad (9)$$

Tunnels

The flow in each tunnel is determined by the difference in water level between start and end and the friction loss factor of the tunnel.

$$\alpha_t q_{n,t,s}^{tunnel} |q_{n,t,s}^{tunnel}| = H_{r_n^{start}}(v_{r_n^{start}}, t, s) - H_{r_n^{end}}(v_{r_n^{end}}, t, s) \quad \forall e \in E, t \in T, s \in S \quad (10)$$

Linearization

Equation (10) is both non-linear and non-convex. The current approach to solving this in GoHydro is to linearize the relationship between discharge in the tunnel and volume in surrounding reservoirs as shown in (11). This converts the formulation into a linear and convex one. However, it is not an exact representation of the physical flow, and may allow for backwards flow in the tunnel, even if the start reservoir is at a higher level than the end reservoir.

$$q_{n,t,s}^{tunnel} <= \frac{\frac{\partial H_{r_n^{start}}}{\partial V_{r_n^{start}}} v_{r_n^{start},t,s} - \frac{\partial H_{r_n^{end}}}{\partial V_{r_n^{end}}} v_{r_n^{end},t,s}}{\alpha_t Q_n^{max}} \quad \forall e \in E, t \in T, s \in S \quad (11)$$

Decomposition

First, we reduce the model complexity by substituting the p- and v-variables out of the problem. For p this is done by inserting the right hand side of equation (2) into the objective function where the p variable is used. For v this is done by inserting the right hand side of equation (5) into inequalities (6), (7) and (11) where the v variable is used.

Next, we decompose the remaining coupling restrictions to solve the model efficiently on a GPU. This is accomplished by relaxing constraints (6), (7) and (11) with Lagrange multipliers and adding them to the objective function. We denote these multipliers $\omega_{r,t,s}^{min}$ for each instance of constraint (6), $\omega_{r,t,s}^{max}$ for each instance of constraint (7) and $\gamma_{n,t,s}$ for each instance of constraint (11).

With these changes, the objective function becomes decomposable for each variable, but in return, each variable will be optimized with respect to the sum of lagrange multipliers and original objective terms related to it. Since the problem is linear, the value of each variable will be either the maximum or minimum allowed value of that variable. Since we have defined the problem as a maximization problem, a variable will be set to its maximum value if the sum of objective coefficients and multipliers is positive. If the sum is negative, the variable will be set to its minimum value.

The master problem of the Lagrange relaxation is to tune the multipliers of relaxed constraints. The goal is to find the value of each multiplier that minimizes the value of the inner maximization problem.

After applying the decomposition, the next equations show the expressions that are valuated to determine the value of each variable.

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